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Improved Resilience by Unlocking Hidden Truths

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Tim Kersley,
Head of Asset Management Strategy,
Network Rail

Network Rail



We own, operate, maintain and develop the railway infrastructure in England, Scotland and Wales.

We connect people and goods with where they need to be and through this support our country's economic prosperity

Our civils infrastructure was largely installed

- between 1850 and 1914 or
- as modernised in the 1950's, 60's and 70's.

Our £500bn, diverse asset base

	Track 32,000km		Bridges 29,000
	Signalling & Level crossings 6,700		Stations and Buildings 2,500
	Electrification 13,000km		Tunnels 800
	Telecoms		Earthworks and Drainage 11,500km

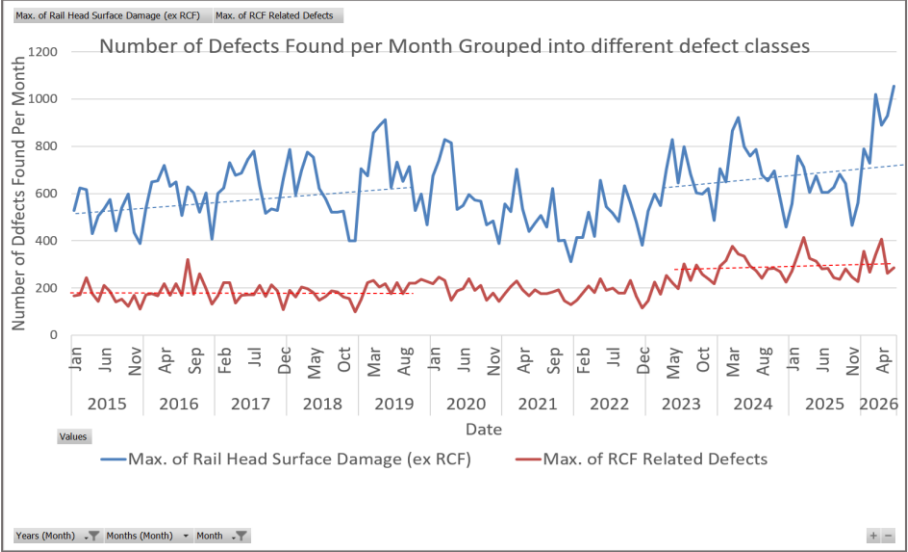
Annual Spend Approximately £3bn Renewal , £1bn Maintenance, £3bn Enhancement

Aging Assets Demand Increasingly Frequent and Accurate Insights

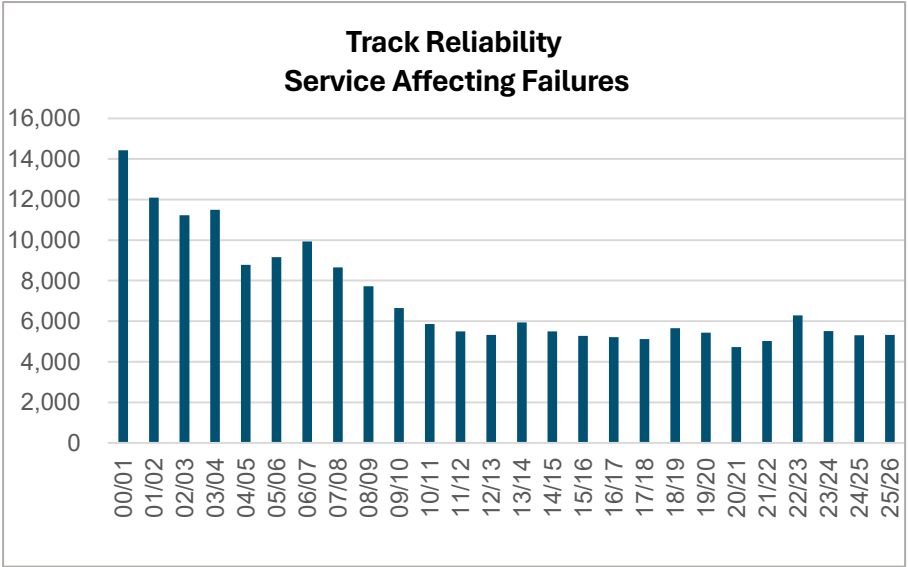
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Asset	% of assets approaching end of life		
	2024	Forecast 2029	Forecast 2034
Track	9.6%	11.4%	12.8%
Signalling	7.1%	9.7%	10.4%
Bridges	13.4%	14.4%	17.3%
Earthworks	4.5%	5.3%	6.3%



Asset	Rate of change in asset approaching end of life	Actionable Defects Growing (last decade)
Track	Yes (50% growth)	Yes (10% growth)
Signalling	Yes (10% growth)	Yes (increased component replacement)
Bridges	Yes (8% growth)	Yes (100% Growth)
Earthworks	Yes (<1% Growth)	Yes (49% Growth)



Track - Proactive Treatments. (10% Life Extension saves £150m / yr)

Insights into rail defects allow proactive management of rail head faults, delivering a greater service life from our legacy assets.

Monitoring capabilities can now be installed on in service trains providing close to real time insights



Technology Advantage

Eddy Current



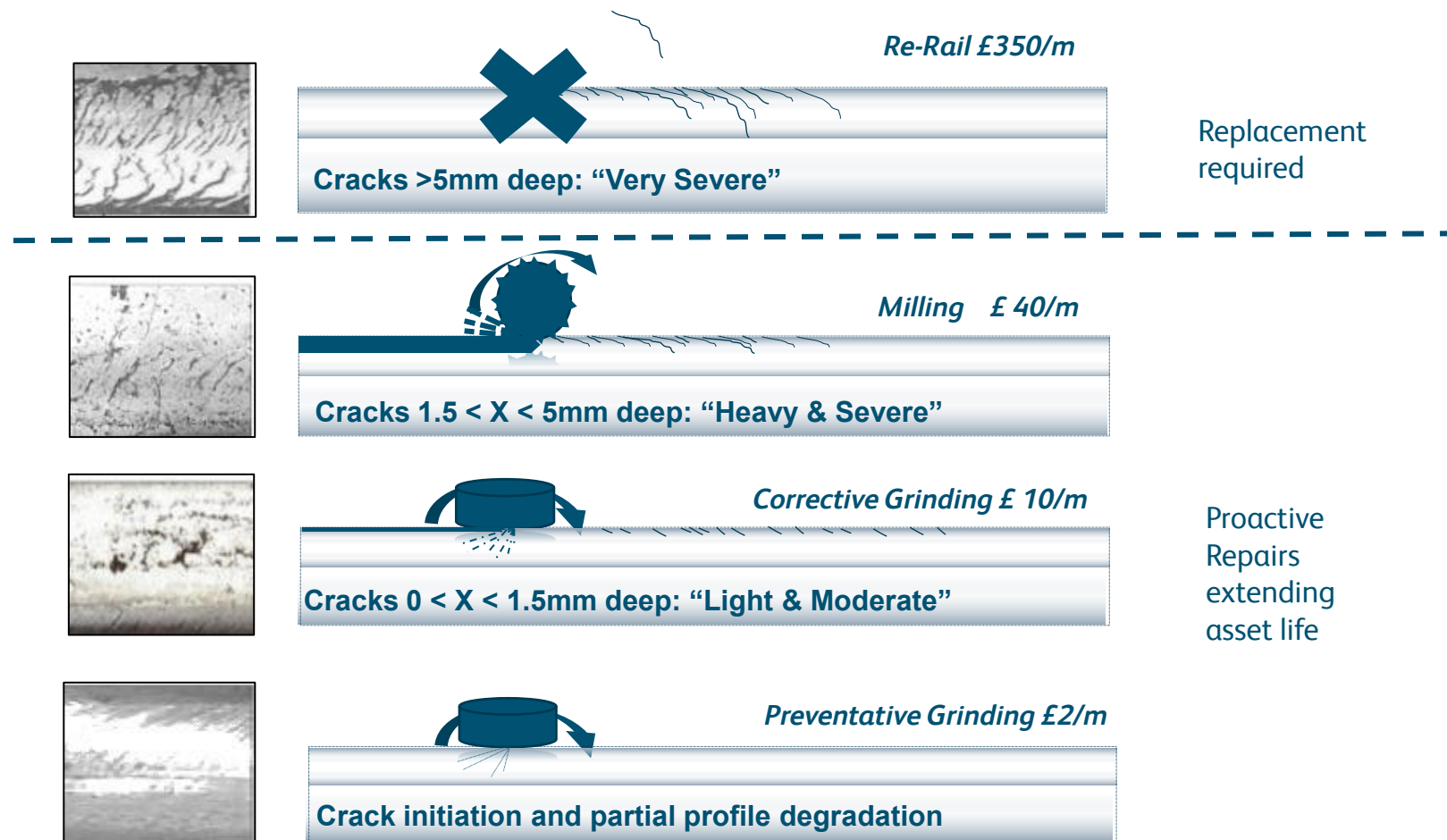
Pattern Recognition



Remote Condition Monitoring



Decision Support Tools (Track Insight)



Overhead Line Equipment

Cost effective, Accurate, Safer, Repeatable

Improved insights extend across other asset types and can again be delivered by in service trains.

PANDAS-V – Pantograph Damage Assessment System - Video

- Video using pattern recognition and AI for analysis.
- Alarms sent to control for investigations.

De-wirements have reduced by approximately 50 %

HotShot

- Thermal imaging camera scanning for signs of overheating and electrical anomalies.
- Cameras that monitor the shoe gear and third rail



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Issue: Contact wire wearing on horn



Issue: Contact wire wearing on horn

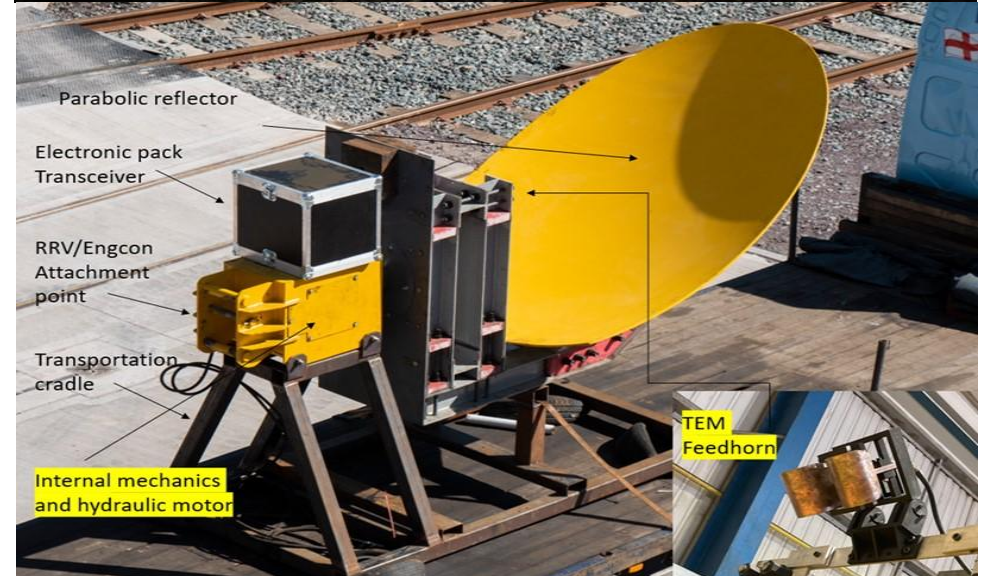
Tunnels: Cost effective, More Accurate , Repeatable , Less disruptive



Automated
Tunnel
Examinations

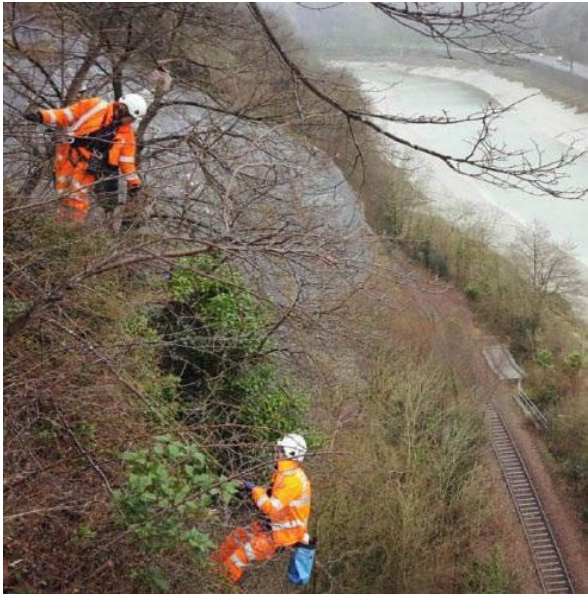


Tunnel
Subsurface
Inspection
Radar (TSIR)

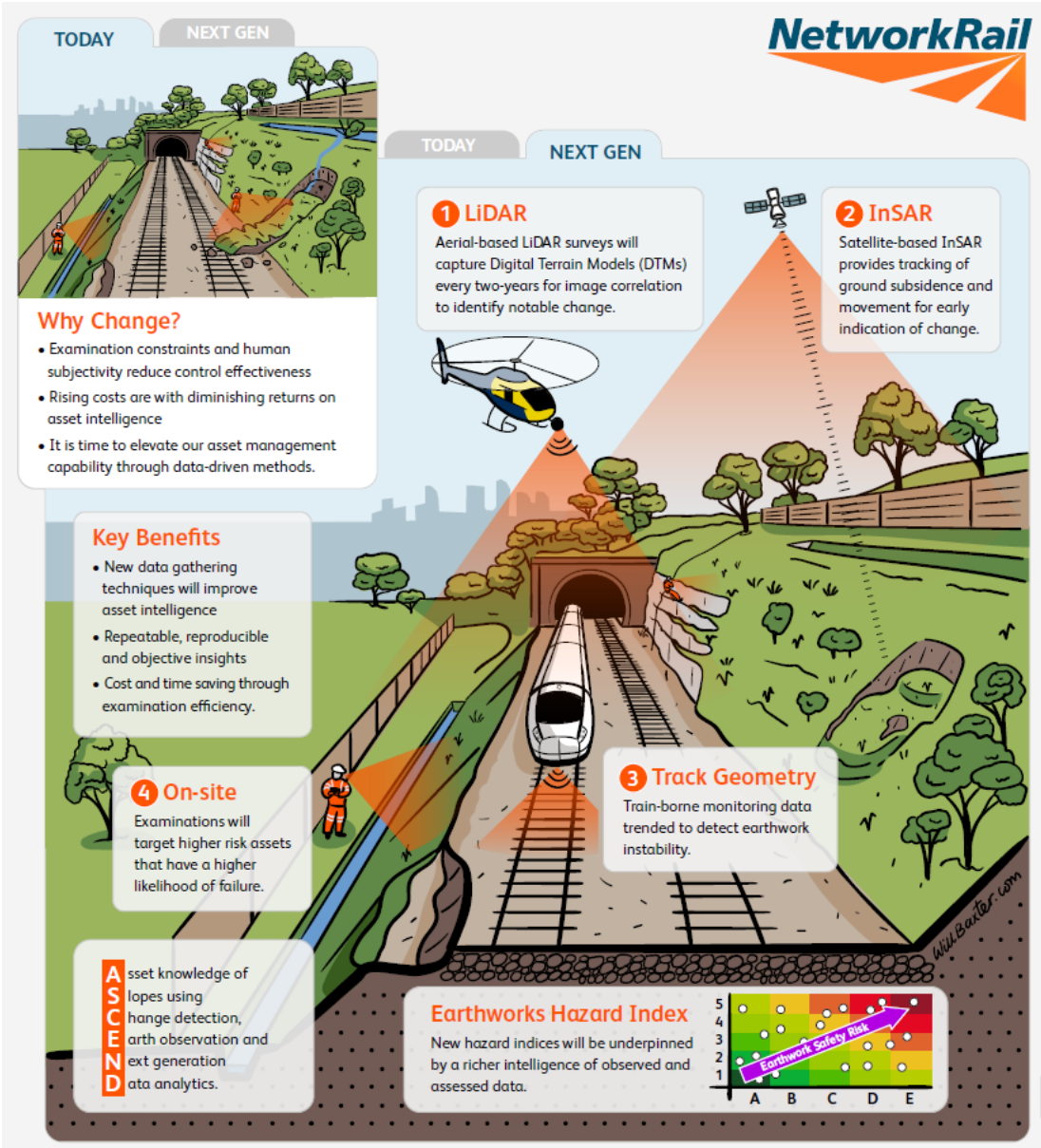
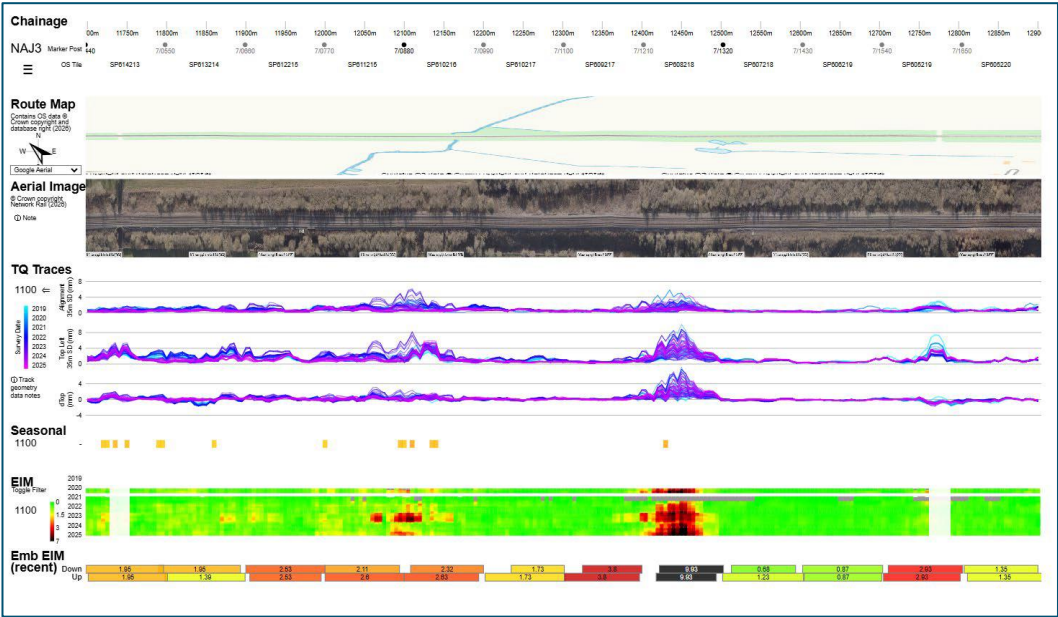


Reveals emerging defects sooner, more reliably.
Early interventions can be planned at lower cost

Earthworks: Cost effective, More Accurate , Repeatable , Less disruptive, Safer



ASCEND
Asset knowledge of
Slopes using
Change detection,
Earth observation and
Next generation
Data analytics.



Summary

Structures: Cost effective, More Accurate , Repeatable , Less disruptive, Safer

Our monitoring capability of our structures lags behind many other asset classes. We need far greater insight to establish the Condition of Hidden Critical Elements

Further development of non-destructive testing, advanced imaging technologies and data integration to unlock the means by which we can target repairs, prevent changes in condition, adopt more durable protective solutions and enhance the resilience and life of bridge infrastructure.

- Detect deterioration
- Applying a rapidly deployable inspection methodology (reduced track possession time)
- Reduce the need for human tactile work (improving workforce safety),
- Use non-intrusive examination methods.
- Include (where necessary) Live remote condition monitoring

Post Tensioned Structures

Learning from Physical tests

Granby Terrace Bridge



- Post-tensioned concrete bridge built in early 1960s, being extended as part of HS2 works.
- Concrete bridge beams were removed and subjected to large-scale testing and forensic demolition.
- Several construction quality defects and subsequent corrosion identified.
- Which of our 375 post tensioned bridges are our greatest risk



Corrosion on
anchorage plate and
nut



Void in grouting



Post Tensioned Concrete Structures

West Coast Main Line reopens after work in Stockport concludes

August 26, 2025



The West Coast Main Line has reopened between Stockport and Staffordshire after a three week closure to enable the demolition and replacement of the Greek Street Bridge.

Greek Street:
£25m

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Replacing these assets is very expensive. The latest M6 crossing cost £63m

Monitoring techniques (like acoustic monitoring) are not only a specialist activity but have proved to be very expensive (costs >£100k for a single bridge).

More readily deployable less disruptive and lower cost monitoring is essential in helping us to target repairs and prioritise replacements

Emerging National Highways work and pilots by TfL is of real interest (muon flux tomography)

What is the future capability to suit an operational railway ?



Clifton Bridge
M62:
£47m